

TURA, Russian Federation

WMO: 245070

Lat: 64.271N

Lon: 100.217E

Elev: 626

StdP: 14.37

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-58.2	-54.5	-60.9	0.1	-57.8	-57.4	0.2	-54.5	12.5	-8.7	10.7	-9.0	0.9	310	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	83.1	63.0	79.1	61.4	75.2	59.3	65.1	78.2	63.2	74.8	61.4	72.0	5.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	80.6	68.4	58.7	75.5	66.8	56.7	70.2	66.0	30.4	78.3	29.0	74.9	27.7	72.3	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.3	9.7	DB	-59.0	88.4	4.3	3.0	-62.1	90.6	-64.6	92.3	-67.0	94.0	-70.1	96.2	(4)
			WB	-59.4	68.0	4.5	1.7	-62.6	69.3	-65.2	70.3	-67.8	71.2	-71.0	72.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	-29.7	-21.1	1.8	21.8	39.2	57.8	62.4	55.2	41.6	20.7	-11.7	-27.5
	DBStd	35.38	15.95	15.97	16.25	11.62	8.91	8.19	6.44	6.46	7.44	13.09	19.33	17.85
	HDD50	12631	2469	1991	1495	845	352	23	1	26	268	908	1851	2402
	HDD65	17324	2934	2411	1960	1295	801	241	129	310	702	1373	2301	2867
	CDD50	860	0	0	0	0	16	258	384	186	16	0	0	0
	CDD65	78	0	0	0	0	0	26	47	5	0	0	0	0
	CDH74	1064	0	0	0	0	9	413	567	74	1	0	0	0
	CDH80	268	0	0	0	0	0	122	139	7	0	0	0	0
Wind	WSAvg	3.7	2.2	2.4	3.6	5.0	5.2	4.5	4.1	3.8	3.9	4.0	2.9	2.3
Precipitation	PrecAvg	14.4	0.6	0.5	0.4	0.6	1.4	1.8	2.0	2.8	1.4	1.3	0.9	0.7
	PrecMax	21.5	1.1	0.9	0.9	1.1	3.2	4.1	4.8	5.6	3.1	2.2	1.9	1.8
	PrecMin	9.6	0.2	0.0	0.1	0.2	0.2	0.4	0.3	0.4	0.1	0.2	0.3	0.1
	PrecStd	3.1	0.2	0.2	0.2	0.3	0.7	0.8	1.3	1.4	0.7	0.6	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	22.4	41.6	52.8	74.8	89.4	88.1	80.7	68.1	47.4	35.7	23.5
		MCWB	11.7	20.2	34.1	41.4	54.2	63.4	66.3	62.9	54.7	41.8	31.9	22.0
	2%	DB	7.2	16.6	35.0	46.8	66.4	83.5	83.8	75.7	62.4	42.8	30.7	15.5
		MCWB	6.1	14.8	30.0	37.5	50.5	62.1	63.7	60.2	52.1	38.5	28.5	14.5
	5%	DB	0.8	11.0	30.5	42.5	59.4	78.7	80.4	71.6	57.6	39.1	25.4	7.4
		MCWB	0.0	9.6	26.3	35.9	46.2	60.2	62.4	58.3	49.8	36.1	23.4	6.4
	10%	DB	-5.3	3.5	26.2	39.0	54.4	74.4	76.8	67.5	53.6	36.0	17.2	0.3
		MCWB	-5.8	2.6	22.9	33.3	43.8	57.6	60.8	57.0	47.7	33.2	15.6	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	20.2	34.6	42.7	55.6	65.8	68.1	66.1	57.3	42.6	32.6	22.1
		MCDB	13.4	22.0	40.5	50.2	72.1	82.2	83.3	77.0	63.2	46.5	35.4	23.4
	2%	WB	6.1	14.8	30.0	38.8	51.4	63.4	65.9	62.5	53.9	39.3	28.8	14.4
		MCDB	7.0	16.5	34.8	45.2	63.5	79.2	80.0	71.3	59.0	41.7	30.9	15.5
	5%	WB	0.0	9.5	26.6	36.3	47.4	61.6	64.2	60.1	51.2	36.3	23.1	6.5
		MCDB	0.7	10.9	30.3	41.6	57.5	75.8	76.0	67.8	56.6	39.0	25.4	7.4
	10%	WB	-5.8	2.7	23.0	33.9	44.2	59.5	62.6	58.3	48.3	33.5	15.8	-0.3
		MCDB	-5.4	3.5	26.4	38.5	53.6	71.5	72.6	65.2	53.2	35.6	17.4	0.3
Mean Daily Temperature Range	5% DB	MDBR	12.1	15.6	24.5	21.2	16.8	20.6	19.8	17.0	14.4	11.5	12.4	12.4
		MCDBR	16.8	17.4	22.6	20.1	26.4	28.9	27.8	26.5	23.2	11.8	13.6	19.8
		MCWBR	16.2	16.1	18.3	13.4	14.1	11.3	11.1	12.2	13.3	8.9	12.6	18.9
	5% WB	MCDBR	16.8	17.3	22.4	18.3	23.6	25.4	22.8	19.7	19.8	11.4	13.9	19.4
		MCWBR	16.3	16.1	18.5	12.8	13.6	10.9	10.1	10.9	12.3	8.9	13.0	19.0
Clear-Sky Solar Irradiance	taub	0.212	0.265	0.299	0.333	0.347	0.369	0.437	0.413	0.330	0.272	0.195	0.199	
	taud	1.876	2.021	2.062	2.067	2.235	2.420	2.199	2.228	2.436	2.223	1.900	1.870	
	Ebn at Noon	100	197	244	263	272	269	242	236	240	198	105	36	
	Edh at Noon	9	21	32	40	37	32	39	35	23	17	8	3	
All-Sky Solar Radiation	RadAvg	54	304	784	1374	1684	1783	1570	1062	650	283	90	20	
	RadStd	5	23	59	90	123	186	144	84	82	35	10	1	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.87	N/A	N/A	+3.17	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)		+1.87	N/A	N/A	+3.17	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page